ENGINEERING DOCUMENTATION PORTFOLIO · VOLUME 07

Runtime Execution and State

Execution lifecycle, state transitions, retries, HITL, and resume.

Eurskem AI · Agentic Workflow Platform

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Derived from the live repository

> **Source of truth.** Inventories and symbol tables are generated from the current checkout. Narrative explains observed structures and does not replace executable contracts.

Execution lifecycle

Run creation establishes owner scope and a durable record. The compiler constructs graph execution. Every node receives resolved configuration and shared services, returns schema-validated output, emits progress, and records status. Failures are classified and persisted. Human gates create resumable state; resume tokens and checkpoints constrain replay.

State classes

- Workflow definition: versioned authoring contract.
- Run document: durable status, inputs, outputs, errors, and timing.
- Graph checkpoint: resumable execution position.
- Event stream: bounded live/replay progress, not canonical history.
- Conversation state: user-facing turns layered over workflow runs.

File	Responsibility
`app/runtime/__init__.py`	Phase 4 runtime: YAML to LangGraph compiler and executor.
`app/runtime/autofix.py`	Auto-fix preflight errors.
`app/runtime/compiler.py`	Compile a WorkflowSpec into a runnable LangGraph StateGraph.
`app/runtime/coordination.py`	Small Redis coordination primitives shared by Uvicorn workers.
`app/runtime/domain_state.py`	Namespaced state for optional use-case packs.
`app/runtime/events.py`	Bounded, session-scoped pub/sub for workflow Server-Sent Events.
`app/runtime/executor.py`	Compile a WorkflowSpec and run it.
`app/runtime/field_schema.py`	Visual field schema — the authoring format behind the Builder's structured output builder.
`app/runtime/hitl.py`	HITL resume path.
`app/runtime/loader.py`	Loader module.
`app/runtime/logic_preflight.py`	Zero-token validation of authored business logic.
`app/runtime/node_events.py`	Helpers for live event emission in the workflow runtime.
`app/runtime/preflight.py`	Zero-token workflow validation.
`app/runtime/rules.py`	Deterministic business-rule evaluation.
`app/runtime/schema.py`	Schema module.
`app/runtime/sleep_guard.py`	macOS-only guard against idle sleep while a workflow is executing.
`app/runtime/snippet_client.py`	App-side client for the snippet-runner sidecar (app/runtime/snippet_daemon.py).
`app/runtime/snippet_daemon.py`	The snippet-runner sidecar's own entrypoint.
`app/runtime/snippet_protocol.py`	Shared shape between the app and the snippet-runner sidecar.
`app/runtime/state.py`	Shared mutable state that flows through every LangGraph node.

File	Responsibility
app/runtime/templating.py	Resolve {{node_id.field.subfield}} expressions inside node configs.
app/workflow/builder_store.py	Durable, file-backed state used by the Workflow Builder.
app/workflow/orchestration.py	Shared execution bookkeeping for a single workflow run.
app/workflow/run_chat_store.py	Durable storage for "Ask AI about this run" conversations.
app/workflow/run_history.py	Durable workflow-run history.
app/workflow/subprocess_callback.py	Deliver a finished child run's result to its waiting Subprocess parent.